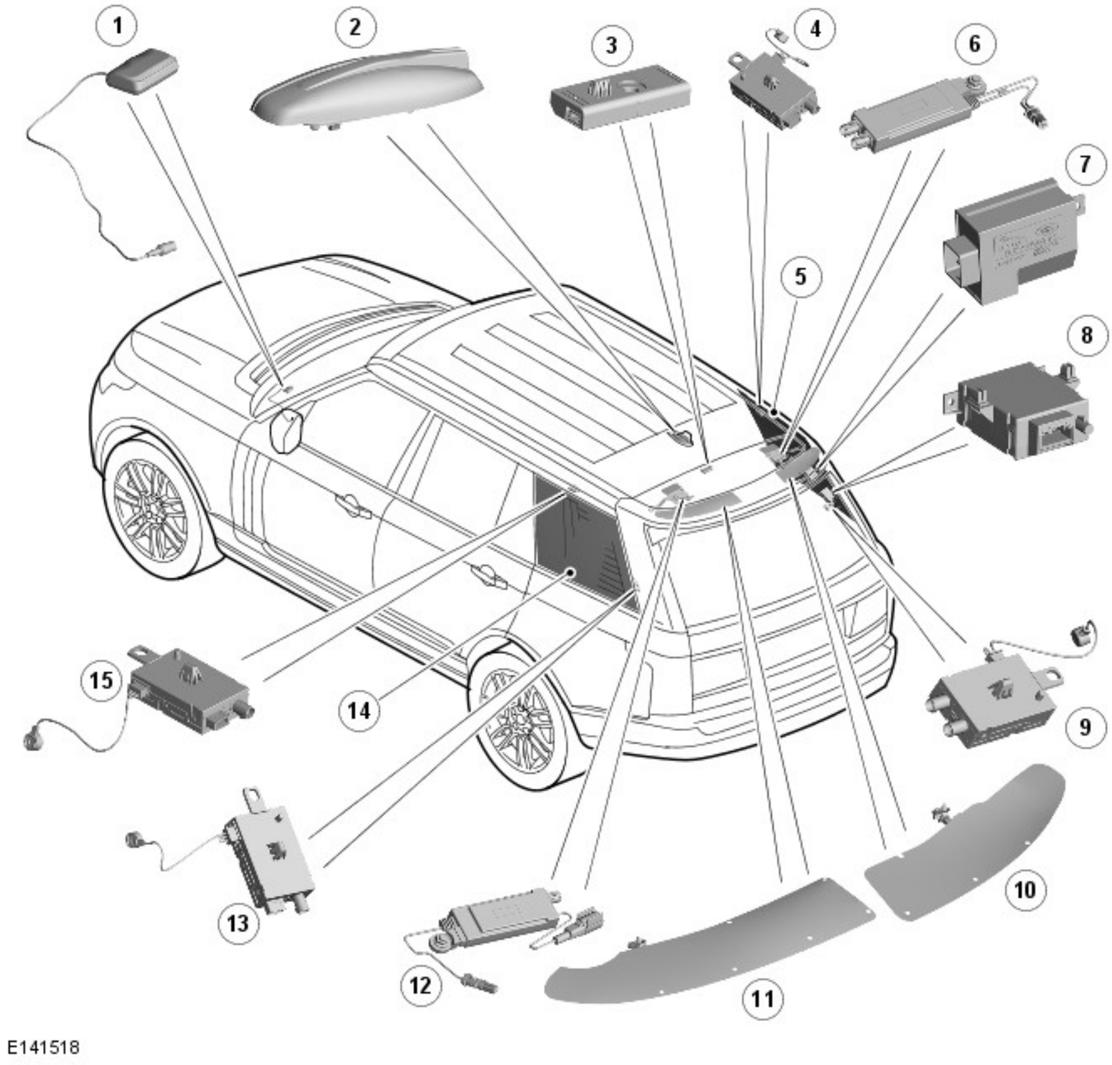
INFORMATION AND ENTERTAINMENT SYSTEM

DESCRIPTION AND OPERATION

ANTENNA - COMPONENT LOCATION



ITEM	DESCRIPTION
1	VICS beacon antenna
2	Roof pod (GPS/GSM/digital radio/satellite radio (NAS)
3	High mounted stop lamp and Heated Rear Window (HRW) RF filter
4	Right upper - TV antenna amplifier
5	Right TV antenna / Fuel Fired Booster Heater (FFBH) remote control antenna print on glass (when fitted)
6	FM2 / digital radio / VICS (Japan only) antenna amplifier
7	Rear wiper motor RF filter
8	FFBH remote control receiver (when fitted)
9	Right lower TV antenna amplifier /FFBH remote control antenna amplifier (when fitted)
10	FM2 / digital radio / TMC / VICS (Japan only) foil antenna
11	AM/FM antenna foil
12	AM / FM antenna amplifier
13	Left lower - TV antenna amplifier
14	Left TV antenna print on glass
15	Left upper - TV antenna amplifier

OVERVIEW

The AM/FM, digital radio band III and VICS (Japan only) antennas foils are located in the rear spoiler. The antennas are part of the spoiler assembly which is a bonded component and therefore the antennas are not serviceable as separate items.

The AM/FM and FM2/DAB/VICS antenna amplifier assemblies are fitted onto the tailgate upper moulding, below the rear spoiler.

Satellite radio (Satellite Digital Audio Radio Service (SDARS)) (NAS only), digital radio Band L, GPS (global positioning system) and Global System for

Mobile communications (GSM) antennas are located in the roof pod.

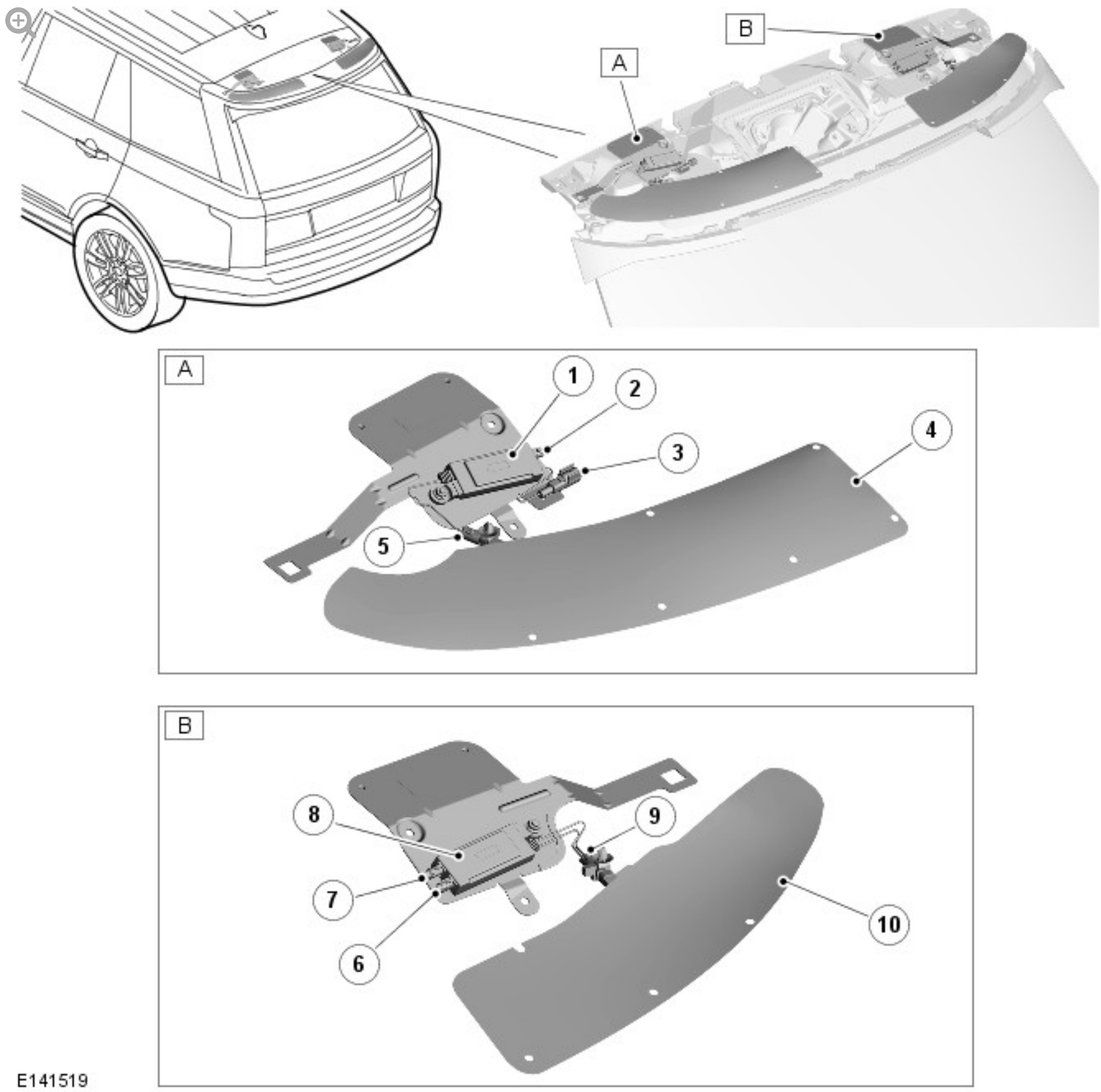
Two foil antennas are located in the rear spoiler which are common to all models.

Two TV antennas are located in each rear quarter window glass and their amplifiers are located above and below the window glass. The TV antenna in the right rear quarter glass is also used for the Fuel Fired Booster Heater (FFBH) remote control RF reception, when fitted.

Japanese specification vehicles can also be fitted with a additional Vehicle Information and Communication System (VICS) beacon antenna.

DESCRIPTION

TAILGATE MOUNTED ANTENNA ASSEMBLIES



ITEM	DESCRIPTION
A	AM/FM Antenna
B	FM2/VICS (Japan only)/Digital Radio (EU/ROW)
1	AM/FM antenna amplifier
2	Connector - Co-axial cable to Integrated Audio Module (IAM)
3	Connector - Power supply from IAM
4	Foil antenna - AM/FM
5	Connector - antenna to amplifier
6	Connector - FM co-axial cable to IAM
7	Connector - DAB or VICS co-axial cable to Navigation control module (Japan only)
8	FM2/VICS (Japan only) or DAB B3 antenna amplifier

9	Connector- antenna to amplifier
10	Foil antenna – FM2/DAB/VICS

AM/FM ANTENNA

The AM/FM antenna is located in the left side of the spoiler. The antenna is a foil type with a single connection to the AM/FM antenna amplifier.

The amplifier is located on a bracket which is attached to the upper tailgate structure.

The amplifier is connected via a co-axial cable to the Integrated Audio Module (IAM). The IAM supplies a power output to the amplifier.

FM2/DIGITAL RADIO BAND III/VICS ANTENNA

The FM2 antenna is located in the right side of the spoiler. The antenna is a foil type with 2 connections to the FM2 amplifier. Depending on market, the amplifier is also utilized for reception of digital radio band III and VICS (Japanese) transmissions.

The amplifier is located on a bracket which is attached to the upper tailgate structure.

The amplifier is connected via 2 co-axial cables to the IAM; one supplying the FM and VICS transmissions and the other supplying the digital radio band III.

VICS /FM2 ANTENNA (JAPAN ONLY)

Data messages for VICS are received through the VICS/FM2 antenna and antenna amplifier located in right side of the rear spoiler. The antenna is a foil type with 2 connections to the amplifier.

The amplifier is located on a bracket which is attached to the upper tailgate structure.

The amplifier has two output connections; one is connected via a co-axial cable to the IAM for FM2 transmissions, the second connection is for the VICS antenna and is connected to the Navigation module via a co-axial cable.

ROOF POD

E141520

Depending on vehicle specification the roof pod can contain GPS, Global System for Mobile communications (GSM), digital radio L-Band or satellite radio (SDARS) antennas.

The roof pod is fitted externally on the roof panel and is attached to the roof panel with two flanged nuts which screw onto captive studs in the roof pod. Two co-axial connectors provide for the connection for the GPS antenna and the satellite or digital radio antenna.

On vehicles from 14MY with the Telematics system fitted, the GPS antenna is connected to a GPS antenna splitter. The splitter has two outputs; one to the Integrated Audio Module (IAM) and one to the Telematics Control Module.

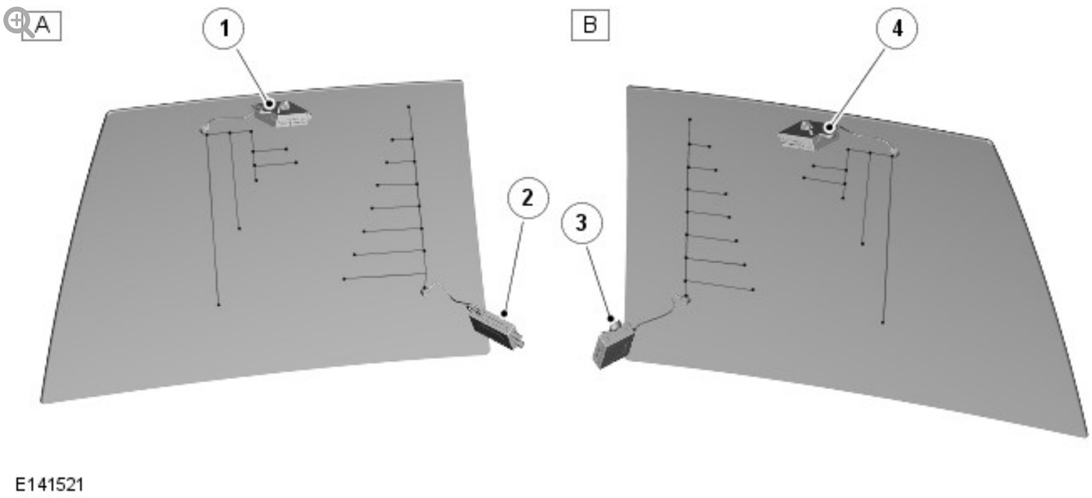
VICS BEACON ANTENNA (JAPAN ONLY)

The VICS beacon antenna is located on the top of the instrument panel. The antenna is attached to the surface of the instrument panel with an adhesive pad.

The VICS beacon antenna is connected to the Navigation control module.

The antenna receives infrared and RF signals from roadside transmitters. The RF reception from the beacon is in addition to the RF reception from the rear spoiler mounted antenna.

TELEVISION (TV) ANTENNAS

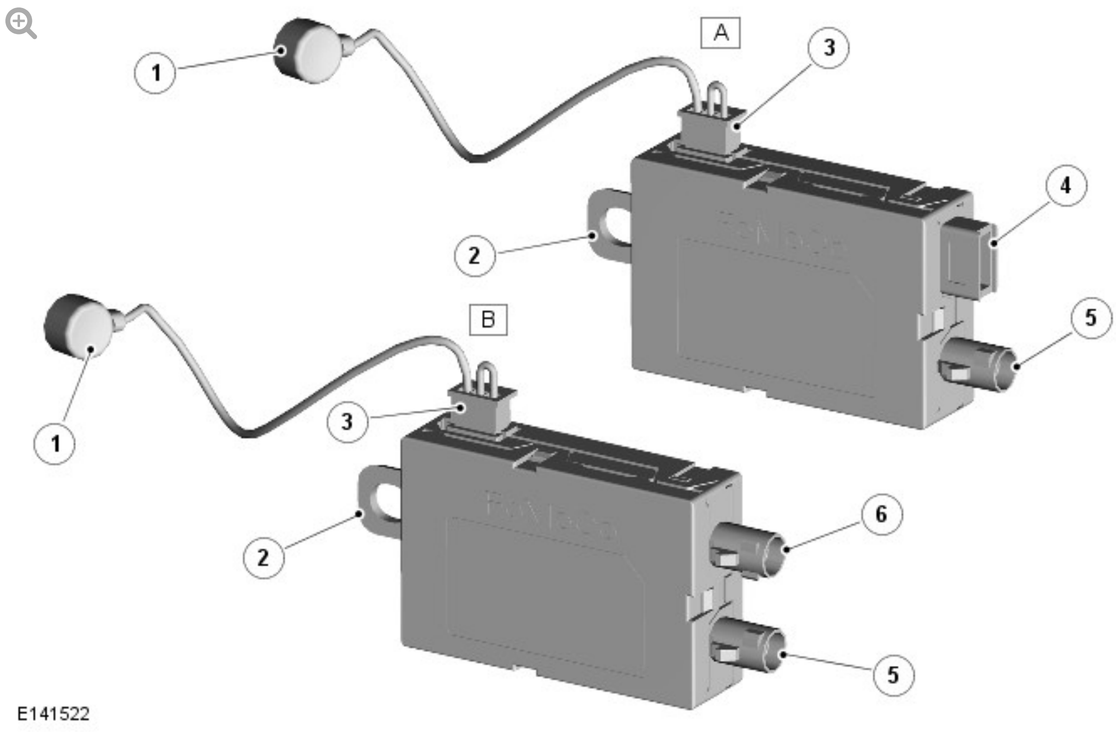


ITEM	DESCRIPTION
A	Right rear quarter window
B	Left rear quarter window
1	TV antenna amplifier
2	TV antenna amplifier/FFBH amplifier (where fitted)
3	TV antenna amplifier
4	TV antenna amplifier

Four TV (television) antennas are used; 2 in each rear quarter window.

The TV/Telestart antenna pattern is printed onto the rear quarter window glass inside surface. Each antenna has a press stud type connector for the connection to the amplifier and each antenna has one amplifier. The lower amplifier on the right side quarter window also controls the RF reception for the Fuel Fired Booster Heater (FFBH) remote control.

TV ANTENNA AMPLIFIERS



ITEM	DESCRIPTION
A	TV only amplifier
B	TV and FFBH remote control RF amplifier
1	Antenna connection
2	Attachment lug/ground
3	Co-axial cable connector to TV control module
4	Co-axial cable connector to FFBH receiver

There are 3 TV amplifiers and 1 TV/FFBH remote control RF amplifier fitted. The antenna on the left side rear quarter window are for TV only. The upper antenna in the right side rear quarter window is for TV only, the lower antenna is for TV and FFBH RF reception.

The 3 TV antenna amplifiers are functionally identical assemblies, but, have handed RF input connections to the press stud on the rear quarter window glass and steering pins. The combined TV/FFBH RF amplifier is a unique unit with 2 co-axial plugs.

A co-axial cable from each amplifier is connected to the TV control module. The FFBH RF signal is passed from the combined amplifier to the FFBH via a co-axial cable.

For additional information, refer to: Fuel Fired Booster Heater (412-02B Auxiliary Climate Control - SDV6 3.0L Diesel - Hybrid Electric Vehicle, Description and Operation).

RADIO FREQUENCY (RF) FILTERS

Two RF filters are used on the antenna system to remove any electrical noise interference being detected by the antenna systems.

A rear wiper motor relay RF filter is located on the right E pillar. The filter is located in the power supply from the Engine Junction Box (EJB) to the rear wiper motor.

A high mounted stop lamp and Heated Rear Window (HRW) RF filter is mounted in a central position on the rear roof cross-member. The filter is located in the power supply from the Central Junction Box (CJB) to the high mounted stop lamp. One of the terminals is also connected to the HRW power feed and filters any RF noise on the power supply wire.

OPERATION

DIAGNOSTICS

The Integrated Audio Module (IAM) can detect the connections between the tuners in the IAM and the antenna amplifiers. Faults in these connections will store a DTC (diagnostic trouble code) in the IAM.

The TV control module can detect the connections between the tuners in the TV control module and the antenna amplifiers.

The connections from the rear quarter window TV antenna elements and the radio spoiler mounted foils and the antenna amplifiers are not monitored by the IAM or the TV control module and therefore a DTC will not be recorded for a fault in these connections. These connections require a manual check of the security of the connectors and a continuity check to ensure there are no broken connections or wires.

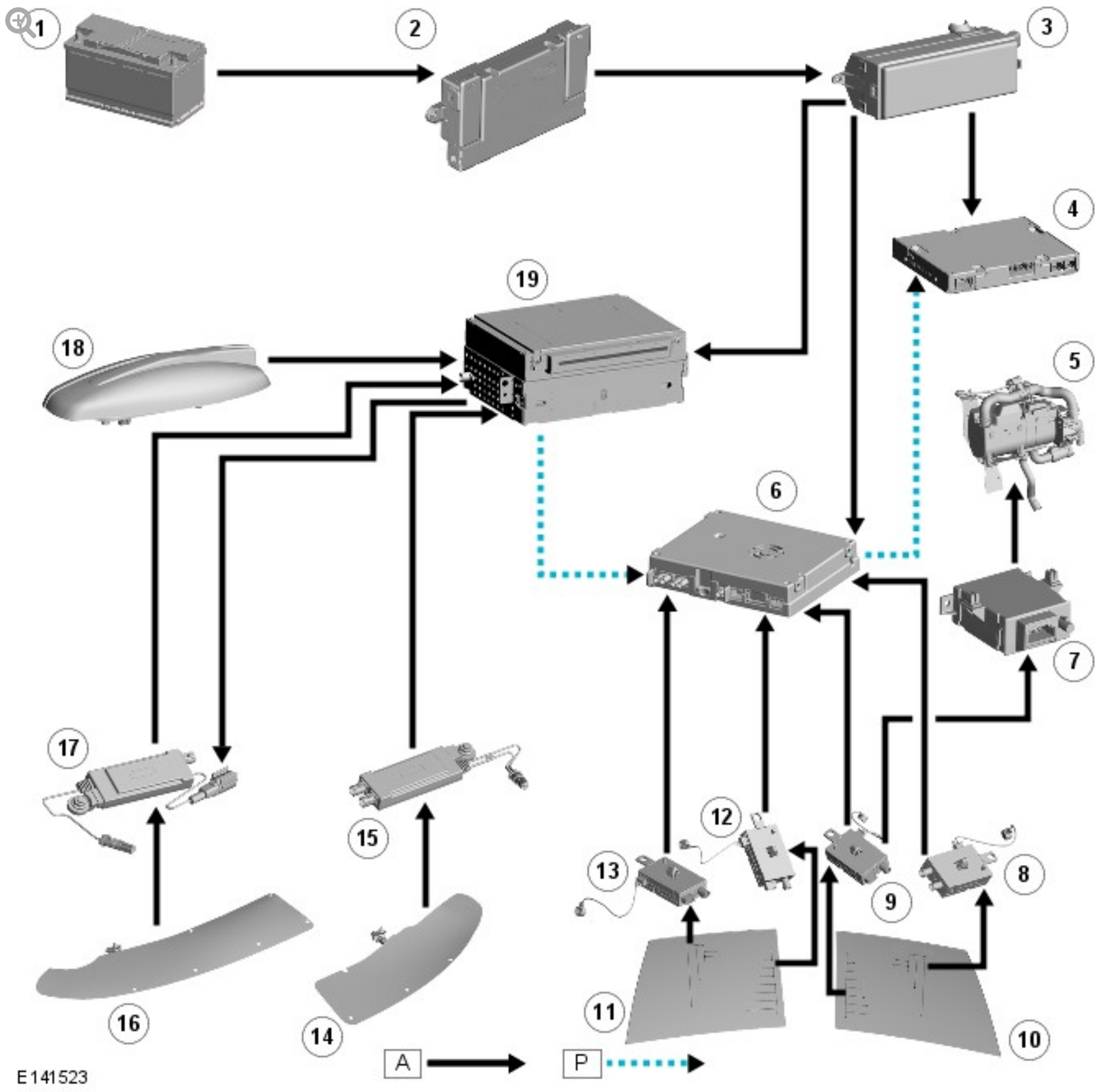
NOTE:

> The foil antennas in the rear spoiler are not serviceable items and replacement requires a new spoiler to be fitted.

The satellite radio (NAS only), the digital radio L-band and the GPS antennas are monitored by their respective control modules and faults in these antennas will record DTCs in the control modules.

CONTROL DIAGRAMS

Control Diagram - Rest of World (ROW)

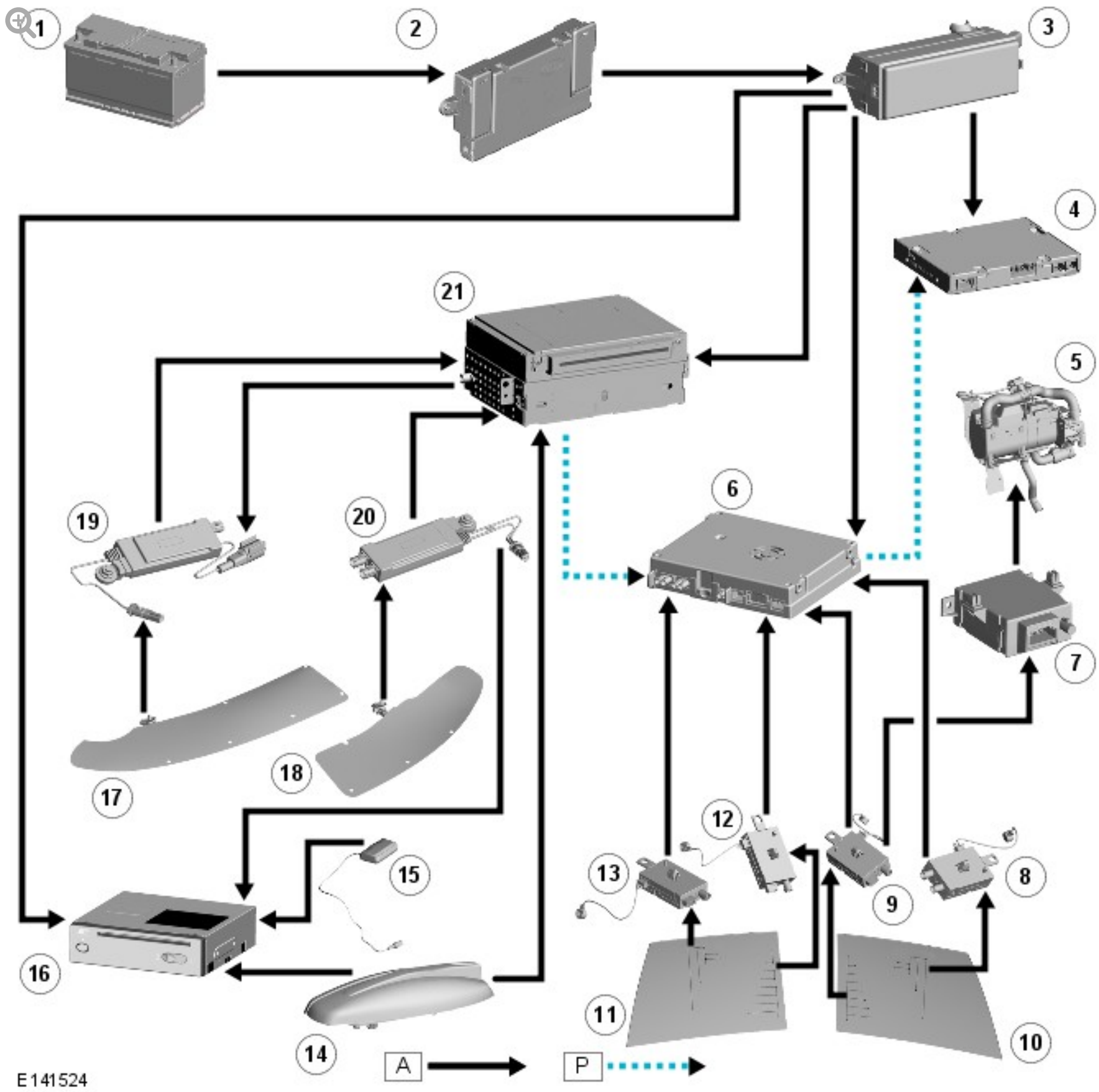


A = HARDWIRED; P = MOST

ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box (BJB)
3	Rear Junction Box (RJB)
4	Rear Seat Entertainment (RSE) control module (where fitted - reference only)
5	Fuel Fired Booster Heater (FFBH) (when fitted - reference only)
6	Television (TV) Control Module
7	FFBH receiver
8	TV antenna amplifier

9	TV antenna amplifier /FFBH remote control antenna amplifier (when fitted)
10	Right TV antenna / Fuel Fired Booster Heater (FFBH) remote control antenna (when fitted)
11	Left TV antenna
12	TV antenna amplifier
13	TV antenna amplifier
14	FM2 / digital radio / TMC foil antenna
15	FM2 / digital radio / TMC antenna amplifier
16	AM / FM foil antenna
17	AM / FM antenna amplifier
18	Roof pod (GPS/GSM/digital radio/satellite radio (NAS))
19	Integrated Audio Module (IAM)

Control Diagram - Japan, Asia



A = HARDWIRED; P = MOST

ITEM	DESCRIPTION
1	Battery
2	Battery Junction Box (BJB)
3	Rear Junction Box (RJB)
4	Rear Seat Entertainment (RSE) control module (when fitted - reference only)
5	Fuel Fired Booster Heater (FFBH) (when fitted - reference only)
6	Television (TV) Control Module
7	FFBH receiver
8	TV antenna amplifier

9	TV antenna amplifier /FFBH remote control antenna amplifier (when fitted)
10	Right TV antenna / Fuel Fired Booster Heater (FFBH) remote control antenna (when fitted)
11	Left TV antenna
12	TV antenna amplifier
13	TV antenna amplifier
14	Roof pod (GPS/digital radio)
15	VICS beacon antenna (Japan only)
16	Navigation control module
17	AM / FM foil antenna
18	FM2 / digital radio / TMC / VICS (Japan only) foil antenna
19	AM / FM antenna amplifier
20	FM2 / digital radio / TMC / VICS (Japan only) antenna amplifier
21	Integrated Audio Module (IAM)